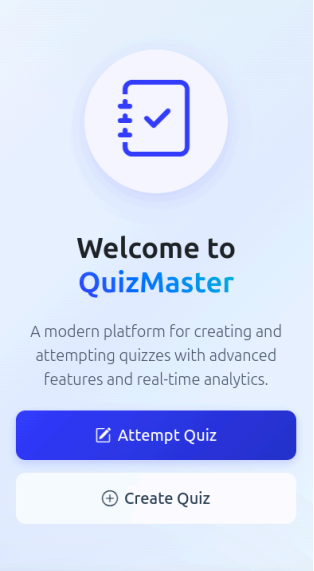
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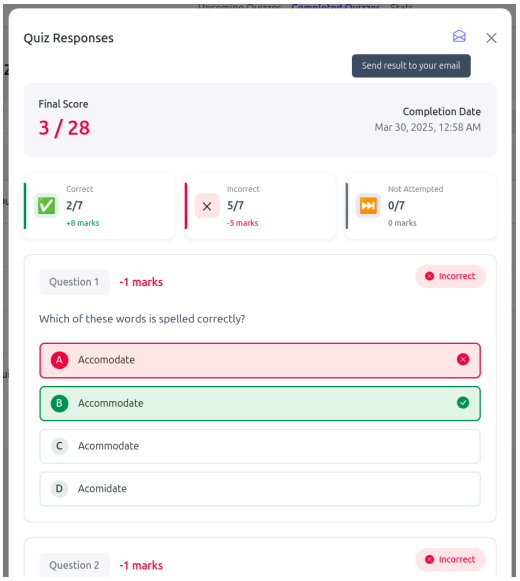
Shahbaz Ali
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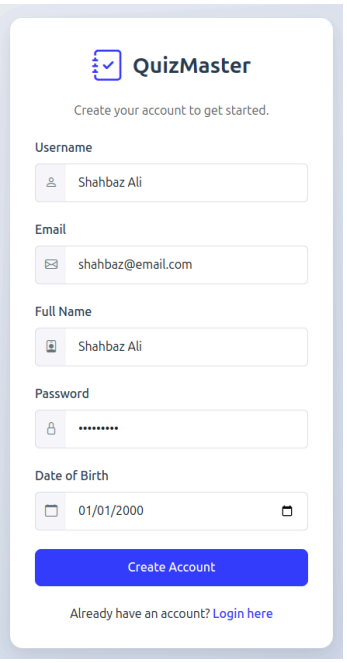
I'm super passionate about coding and building things. I love solving problems, exploring new technologies, and creating projects that make an impact. Whether it's designing databases, optimizing APIs, or experimenting with new frameworks, I enjoy turning ideas into reality.

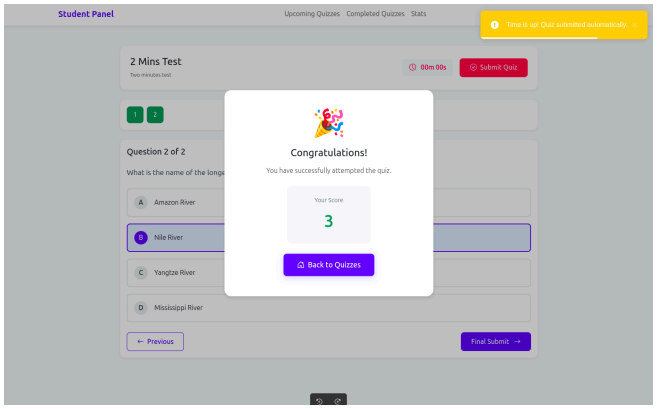
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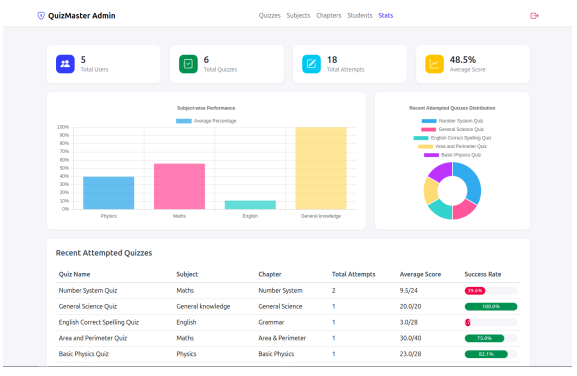
QuizMaster is an AI-powered quiz management system that allows educators to generate and manage quizzes while enabling students to attempt quizzes, track progress, and view performance analytics. The project involves building a responsive web interface with automated question generation, real-time assessments, and interactive learning features.











Technologies used

- Backend

Flask	Python web framework for building the REST API
Flask-SQLAlchemy	ORM for database operations
Flask-JWT-Extended	JWT authentication and authorization
Flask-Migrate	Database migration management
Bcrypt	Password hashing
SQLite	Lightweight, serverless database for data storage
Redis	In-memory data store for caching.
Celery	Distributed task queue for background jobs
reportlab	PDF report generation
Google-generativeai	AI-powered question generation

- Frontend

Vue.js	Progressive JavaScript framework
Bootstrap	CSS framework for responsive design
Bootstrap Icons	Icon library
Vue Toastification	Toast notifications
Chart.js	Charting library

DB Schema Design

The database schema is designed with flexibility and extensibility in mind, particularly in handling different question types.One key design choice was creating a wrapper table for questions, encompassing MCQ and Numeric types. This structure allows seamless support for multiple question formats, such as integer, numeric, and subjective, with minimal changes, making the system easily extensible for future question types.



Fig: ER Diagram

API Design

The QuizMaster API is built using Flask and implements RESTful endpoints for user management, quiz operations, and analytics. Key API groups include:

1. **Auth** : APIs for Login, registration
2. **Admin**
    a. **Subject**: Subject CRUD Apis
    b. **Chapter**: Chapter CRUD APIs
    c. **Quiz**: Admin APIs to manage quiz
    d. **Attempts**: APIs to fetch student attempts and quiz responses
3. **Student**
    a. **Attempts**: APIs to start/end quiz, fetch attempt and quiz response

Postman API Doc: https://documenter.getpostman.com/view/32768981/2sB2cPk5ny

Architecture and Features

The project follows a modular architecture with clear separation of concerns between frontend and backend components. The backend is organized using Flask's blueprint pattern, with controllers (routes) in `backend/app/routes/`, models in `backend/app/models/`, and utilities in `backend/app/utils/`. The frontend follows Vue.js

best practices with components in `frontend/src/components/`, views in `frontend/src/views/`, and shared utilities in `frontend/src/utils/`. Configuration files are separated by environment (development/production) and stored in their respective directories. The project uses a service-based architecture where business logic is encapsulated in service classes, making the codebase maintainable and testable.

Features Implemented:

- ✅ AI powered question generation from context.
- ✅ Auto DB initialization, migration from code
- ✅ JWT and Role based authentication
- ✅ CRUD for Subjects and Chapters
- ✅ Admin Quiz creation, updation and deletion
- ✅ Two type of questions, MCQ and Numeric
- ✅ Question Creation, Updation and Deletion by admin
- ✅ Admin Stats page
- ✅ Chart JS for displaying charts
- ✅ Admin automated monthly pdf report sending on email
- ✅ Student Email notification on inactivity
- ✅ Student Email notification when new quiz is created
- ✅ Redis for caching
- ✅ Celery for asynchronous processing of background tasks like email sending
- ✅ Frontend validation of forms
- ✅ Backend validation of APIs
- ✅ Responsive UI for Desktop, Tablets and phones
- ✅ Filter quizzes based on subject and chapter.
- ✅ Paginated APIs
- ✅ Install the website as a PWA app.
- ✅ Quiz auto submission
- ✅ Optional Negative marking
- ✅ Detailed quiz response sheet.
- ✅ Student Statistics page.
- ✅ Quiz result export in csv format.

Video

quiz_master_v2_video.mp4